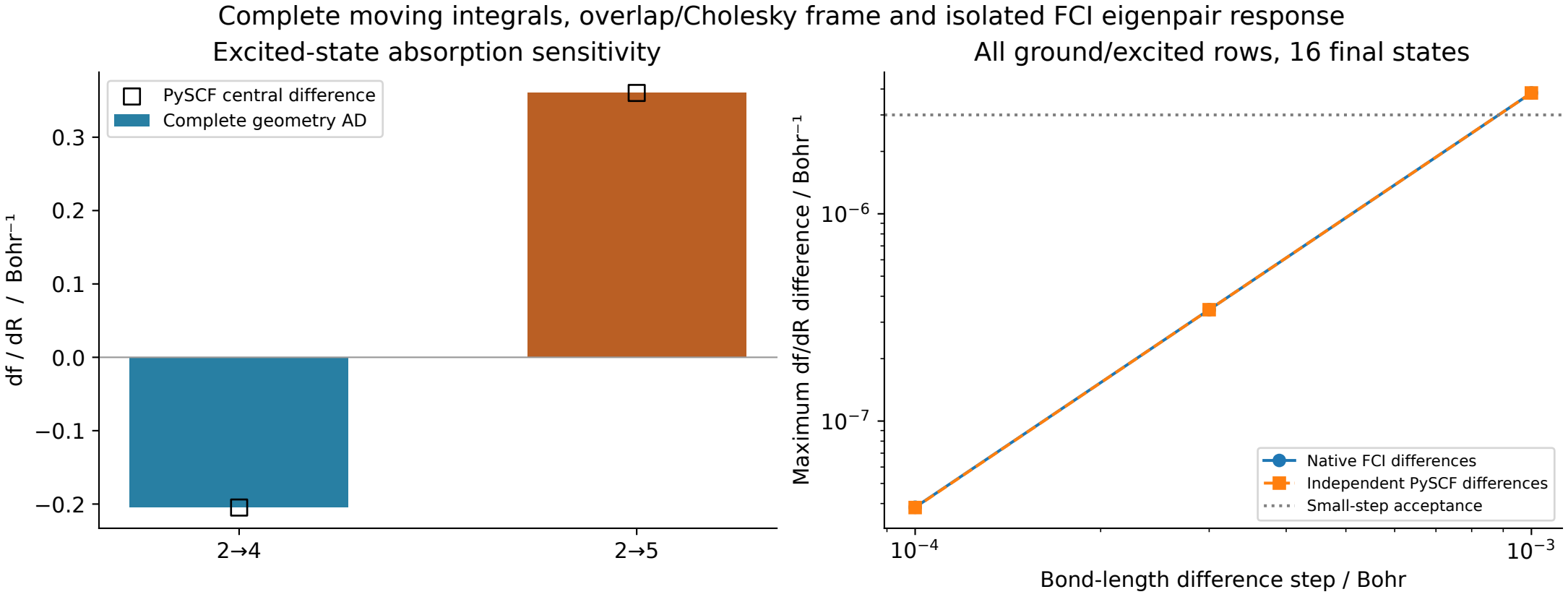
Complete moving integrals, overlap/Cholesky frame and isolated FCI eigenpair response
Excited-state absorption sensitivity
PySCF central difference
Complete geometry AD
df / dR / Bohr⁻¹
0.3
0.2
0.1
0.0
−0.1
−0.2
2→4
2→5
All ground/excited rows, 16 final states
Maximum df/dR difference / Bohr⁻¹
10⁻⁶
10⁻⁷
Native FCI differences
Independent PySCF differences
Small-step acceptance
10⁻⁴
10⁻³
Bond-length difference step / Bohr